The proposed courseware and outline

Using Data to Drive Business Performance

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The proposed courseware

One connected learning journey: from understanding a business question to building the database, analyzing the evidence and making a decision.

An invitation to develop this together
A working proposal, open to better ideas.This outline is a draft and a suggestion, informed by the research and my classroom experience. I would like to work with the editors and reviewers to finalize the sequence, depth, tools and running projects. It is a starting point for our conversation.

Read, build, try, reflect

Each chapter introduces a business question, develops the core concepts, and includes a guided **Let's Build** , review and reflection questions, a **Reading Assessment Test** , and an independent **interactive lab** . Terms and takeaways support review.

A project that grows with the student

Two running cases connect the chapter activities. A separate semester project moves through a proposal, a working database and a final dashboard with a decision memo. Review the cases ↗ · See the assessment plan ↗

The complete chapter outline

Sixteen chapters in five parts, with a preface, a midterm integration studio, a closing chapter on careers, and seven appendices. Every chapter section is visible below, followed by the guided practice, independent work and research connection.

Preface · Preface: Why This Book, and How It Works

Why it was written and how to use it.

• P.1 Why I Wrote This Book

• P.2 How This Book Is Different

• P.3 How a Chapter Works

• P.4 The Tools, and Why They Were Chosen

• P.5 Two Routes Through the Book

• P.6 How AI Is Used in This Book

Part I · Chapters 1 to 3

From Data to Decisions

Main goal: Connect data to business performance, classify data correctly and organize information.

Chapter 1 · From Data to Business Performance

How does a business turn everyday activity into better results?

• 1.1 What a Business Is and How It Creates Value

• 1.2 Business Performance, Goals and KPIs

• 1.3 From Data to Wisdom: The DIKW Pyramid

• 1.4 Information Behavior: How People Seek and Use Information

• 1.5 Information Systems and Information Technology

• 1.6 The R.E.A.D. Framework: Recognize, Extract, Analyze, Decide

• 1.7 AI in the Data-to-Decision Chain

• 1.8 The Running Cases and the Semester Project

Let's Build: Trace one KPI back to the records that produce it.

Interactive lab: A KPI audit of a small business: three KPIs, the data each needs, and who decides with it.

Research connection: None of the four compared texts opens with business performance and KPIs. Business-systems framing appears in Buffalo MGS 351, Syracuse IST 195 and UNLV MIS 766.

Chapter 2 · Understanding Data

What kind of data is this, and what can I legitimately do with it?

- 2.1 Data and Information
- 2.2 Digital and Analog Data; Bits, Bytes and Tokens
- 2.3 Data Types: Numbers, Text, Dates, Booleans and Files
- 2.4 Qualitative and Quantitative Data
- 2.5 Levels of Measurement: Nominal, Ordinal, Interval, Ratio
- 2.6 Structured, Semi-Structured and Unstructured Data: CSV and JSON
- 2.7 What Makes a Good Table
- 2.8 Data Quality
- 2.9 From Data to Insight: What Each Type Allows

Let's Build: Profile a dataset column by column: type, scale and allowed operations.

Interactive lab: Data Type Detective: classify and repair a messy business file, then draw one valid insight per column type.

Research connection: The compared outlines do not give data literacy and measurement scales their own opening chapter. Data preparation appears in NYU TECH-UB 26 and Temple MIS 2502.

Chapter 3 · Organizing Data and Information

How do we make data findable, reusable and trustworthy?

- 3.1 Why Organization Matters
- 3.2 Files, Folders, Naming and Versions
- 3.3 Hierarchies, Tags and Facets

• 3.4 Metadata, with YAML as an Example

• 3.5 Taxonomies and Controlled Vocabularies

• 3.6 Data Dictionaries

• 3.7 Spreadsheets as Tables, and Where They Break

• 3.8 From Organized Files to Databases

Let's Build: A data dictionary and a folder-and-tag scheme for the running case.

Interactive lab: Reorganize and document a shared-drive mess.

Research connection: Added by design from information science, so students see why databases exist before they build one. Not a standalone topic in the syllabi reviewed.

Part II · Chapters 4 to 6

Databases and SQL

Main goal: Read a relational database and answer business questions with SQL.

Chapter 4 · Databases and the Relational Model

Why do organizations keep data in related tables instead of files?

• 4.1 What a Database Is and What a DBMS Does

• 4.2 Local and Cloud Databases: SQLite, PostgreSQL, Supabase, MySQL, Access

• 4.3 Tables, Rows, Columns and Data Types

• 4.4 Primary, Foreign and Composite Keys

• 4.5 Relationships: One-to-One, One-to-Many, Many-to-Many

• 4.6 Schemas, Integrity Rules and Constraints

• 4.7 Database Objects: Tables, Queries, Views, Forms and Reports

Let's Build: Explore a supplied case database in Supabase Studio and trace a relationship.

Interactive lab: Read an unfamiliar schema and predict the answers to five business questions before querying.

Research connection: Design and data modeling are named in all 13 catalogue benchmark descriptions; relational foundations are documented in Baruch CIS 3400, George Mason MIS 310 and SUNY Canton CITA 215.

Chapter 5 · SQL Foundations

How do I ask one table a precise question?

- 5.1 SELECT and FROM
- 5.2 Filtering with WHERE, LIKE, IN and BETWEEN
- 5.3 Sorting, DISTINCT and LIMIT
- 5.4 Working with NULL and Dates
- 5.5 Aggregate Functions
- 5.6 Creating Tables and Inserting Data
- 5.7 Safe UPDATE and DELETE
- 5.8 AI verification: Accept, Revise or Reject an AI-Written Query

Let's Build: Answer ten business questions on the running case.

Interactive lab: The same skills on the second case, with an AI verification.

Research connection: SQL appears in 10 of 13 catalogue benchmark descriptions (77%) and in 52.9% of the 855 analyst postings. Early SQL in Rutgers 33:136:470, Northeastern CS 3200 and UT Dallas ITSS 4300.

Chapter 6 · SQL for Business Analysis

How do I combine tables into trustworthy business measures?

- 6.1 INNER and LEFT Joins
- 6.2 Joining Three or More Tables
- 6.3 GROUP BY and HAVING
- 6.4 CASE Expressions for Business Categories
- 6.5 Calculated Measures: Rates, Averages and Shares

• 6.6 Grain and the Double-Count Problem

• 6.7 Checking Results by Hand

Let's Build: Build a small KPI report from joined tables.

Interactive lab: Find and fix a double-counted measure.

Research connection: Joins and aggregation are where business reports most often go wrong. Taught at Rutgers, Temple, UT Austin and UT Dallas.

Midterm studio · Integration Studio A (midterm)

An unnumbered review week between Parts II and III.

Cumulative review of Chapters 1 to 6, a practice exam, and **Project Checkpoint 1** : proposal and data plan.

Research connection: Staged, multi-phase projects recur across the course documents reviewed, so the project has graded checkpoints through the semester.

Part III · Chapters 7 to 11

Designing and Running Databases

Main goal: Design, normalize, build, test and secure a database of their own.

Chapter 7 · Database Design: From Requirements to ER Diagrams

How do business rules become a database design?

• 7.1 The Systems Development Life Cycle

• 7.2 Waterfall and Agile

• 7.3 Requirements and Business Rules

• 7.4 Entities, Attributes and Relationships

• 7.5 Cardinality, Optionality and Crow's Foot Notation

• 7.6 Associative Entities and Many-to-Many Relationships

• 7.7 Drawing ERDs with Lucidchart, diagrams.net or Mermaid

Let's Build: Draw the ERD for the running case from its business rules.

Interactive lab: Design an ERD for a new business process.

> **Research connection:** Design and data modeling are named in all 13 catalogue benchmark descriptions. ER modeling comes before normalization, as in Coronel and Morris, Kroenke and Hoffer.

Chapter 8 · Normalization

How do we design out errors before they happen?

- 8.1 Update, Insert and Delete Anomalies
- 8.2 Functional Dependencies
- 8.3 First Normal Form
- 8.4 Second Normal Form
- 8.5 Third Normal Form
- 8.6 Decomposing and Rejoining Tables
- 8.7 When to Denormalize

Let's Build: Normalize a flat spreadsheet to third normal form.

Interactive lab: Normalize a real business form and prove no data was lost.

> **Research connection:** Normalization is core in each of the four compared texts and is documented in Baruch, Rutgers, George Mason, Georgia Tech and SUNY Farmingdale BCS 260.

Chapter 9 · Building and Testing a Database

How do we turn a design into a working, trustworthy database?

- 9.1 Data Definition Language: CREATE, ALTER, DROP
- 9.2 Constraints
- 9.3 Loading Data from CSV
- 9.4 Staging and Cleaning
- 9.5 Integrity Tests

• 9.6 Data Entry Workflows and Forms

Let's Build: Build and load the running-case database in Supabase.

Interactive lab: Build the lab case from its ERD and run an integrity test suite.

Research connection: Hands-on building appears in 10 of 13 catalogue benchmark descriptions (77%). Implementation projects at Rutgers, Baruch and Boston University MET CS 669.

Chapter 10 · Advanced SQL

How do analysts write queries that are reusable and easy to check?

• 10.1 Subqueries

• 10.2 Common Table Expressions

• 10.3 Views

• 10.4 Conditional Aggregation

• 10.5 Window Functions

• 10.6 Set Operations

• 10.7 Functions and Triggers

• 10.8 Reading a Query Plan with EXPLAIN

Let's Build: A reusable analysis layer of views and CTEs.

Interactive lab: A monthly performance report with rankings and period-over-period change.

Research connection: Analytical SQL with CTEs and window functions appears in current syllabi at NYU Stern and UT Dallas.

Chapter 11 · Database Administration, Security and Reliability

How do we keep business data safe, correct and available?

• 11.1 Roles, Users and Least Privilege

• 11.2 Row-Level Security

• 11.3 Transactions and ACID

• 11.4 Concurrency

• 11.5 Backup and Recovery

• 11.6 Indexes and Performance

• 11.7 Governance, Privacy and Compliance

• 11.8 Audit Trails

Let's Build: Add roles, row-level security and an audit trigger to the running case.

Interactive lab: A security and recovery plan for the lab case. Project Checkpoint 2: working database.

Research connection: Administration, security or transactions appear in 6 of 13 catalogue benchmark descriptions (46%), including Rutgers, UT Austin, George Mason and SUNY Canton.

Part IV · Chapters 12 to 15

From Data to Insight

Main goal: Analyze results, visualize them, report in Power BI and work with warehouses and AI.

Chapter 12 · Business Analytics and Decision-Making

What does the data actually tell us, and how sure are we?

• 12.1 Four Kinds of Analytics

• 12.2 Framing an Analytical Question

• 12.3 Distributions, Segments and Fair Comparisons

• 12.4 Correlation and Causation

• 12.5 Uncertainty

• 12.6 From Measure to Decision

• 12.7 The Decision Memo

Let's Build: A SQL analysis that answers one management question.

Interactive lab: An analytical memo for the lab case.

Research connection: Closes the loop from query to decision with a decision memo. Business analysis tasks appear in Temple MIS 2502 and Georgia MIST 4610.

Chapter 13 · Reporting and Data Visualization

How do we show results so people decide well?

- 13.1 Reports and Dashboards
- 13.2 Choosing the Right Chart
- 13.3 Honest and Misleading Visuals
- 13.4 Accessible Design
- 13.5 Designing for Decision-Makers
- 13.6 The Tableau Route

Let's Build: Redesign a misleading dashboard.

Interactive lab: A one-page dashboard specification.

Research connection: Only 4 of 13 catalogue benchmark descriptions (31%) include BI or analytics. Visualization appears in Georgia MIST 4610 and UT Austin MIS 325.

Chapter 14 · Power BI

How do we turn a database into an interactive business report?

- 14.1 Connecting to a Database
- 14.2 Shaping Data with Power Query
- 14.3 The Data Model and Relationships
- 14.4 Measures, Calculated Columns and DAX Basics
- 14.5 Building an Interactive Report
- 14.6 Validating Every Number Against SQL

Let's Build: A Power BI report on the running case.

Interactive lab: A Power BI report on the lab case, validated against SQL.

Research connection: Power BI is listed in 29% of the 855 analyst postings and is not documented in any of the course families reviewed, so it is added by design.

Chapter 15 · Big Data, Data Warehouses and AI

How do organizations work with data at scale, and where does AI fit?

- 15.1 OLTP and OLAP
- 15.2 Warehouses, Marts, Lakes and Lakehouses
- 15.3 Star Schemas: Facts and Dimensions
- 15.4 ETL, ELT, APIs and JSON
- 15.5 Big Data and NoSQL
- 15.6 AI and Data: Text-to-SQL, Embeddings and Retrieval
- 15.7 Verifying AI Output

Let's Build: A small star schema from the running case.

Interactive lab: Evaluate three AI answers about the lab data: accept, revise or reject.

Research connection: The 13-course catalogue benchmark names AI in one description and cloud computing in none. Warehousing appears in Baruch CIS 4400 and graduate AI topics in UT Dallas BUAN 6V99; these inform extensions, not universal requirements.

Part V · Chapter 16 and Closing

Putting It to Work

Main goal: Deliver a capstone decision package and plan a path into data careers.

Chapter 16 · Capstone: From Data to Decisions

Can you take a real business problem all the way to a defensible decision?

- 16.1 Scoping the Problem and the KPI
- 16.2 From Data Dictionary to ERD
- 16.3 Building the Database
- 16.4 Analysis and a Validated Dashboard
- 16.5 The Decision Memo
- 16.6 Demonstration and Reflection

Project Checkpoint 3: the final project, with a working database, analysis, dashboard and decision memo, and an individual demonstration.

Research connection: Staged projects at Rutgers, Boston University and Georgia. The individual demonstration responds to the uneven AI policies documented in the syllabi.

Closing chapter · Closing: Where Data Can Take You

Where do these skills lead?

• C.1 The Data-to-Decision Journey on One Page

• C.2 Careers in Data: Analyst, BI Analyst, Database Administrator, Data Architect, Data Engineer

• C.3 Data Skills in Every Business Major

• C.4 Building a Portfolio

Research connection: The labor and employer research above: data scientist jobs growing 35% and database architect jobs 9% through 2035.

Appendices

These companions keep setup steps and platform alternatives close at hand without interrupting the main story. Instructors can choose the tools their course needs, while students follow the same core concepts and cases.

A

Supabase and the In-Book SQL Editor

Setting up a Supabase project, using Supabase Studio, and (planned) running PostgreSQL inside the reader.

B

MySQL Route

Planned: schemas, seed data and solutions for every SQL chapter, to be tested on the shared cases, for courses built on MySQL.

C

Microsoft Access Route

Tables, relationships, visual queries, forms and reports on the same cases. An Access appendix and Access labs exist today.

D

SQL Reference and Dialect Guide

PostgreSQL, MySQL, SQL Server and SQLite side by side.

E

Diagramming with Lucidchart

ER diagrams in Lucidchart, with diagrams.net and Mermaid as free alternatives.

F

Tableau Route

The visualization chapter's activities in Tableau.

G

Aa

Terms Treasury

The glossary of every key term in the book.

How students progress

• **Understand the data** Chapters 1 to 3: what data is, how it measures performance, and how to organize it.

• **Use a database** Chapters 4 to 6: query a supplied database before designing one.

• **Build one** Chapters 7 to 11: design, normalize, build and secure their own database.

• **Decide and defend** Chapters 12 to 16: turn results into dashboards and a decision memo they can explain and defend.

Two ways to use the book

Both routes reflect the syllabus research: business-core courses emphasize data use, design and reporting, while MIS courses need the full database sequence. Both are selectable; final pacing is to be agreed in review.

Business-core data course	Chapters 1 to 8, then 12 to 14.
MIS database course	Chapters 1 and 4 to 16, with Chapters 2 and 3 as assigned reading.

Technology and tools

The syllabus platform audit distinguishes documented practical use, a named option, and a software-list mention. Counts use 54 selected course families and overlap across tools. Official documentation then informs the proposed PGlite, Supabase, and Power BI pathway. [S2] [T1] [T2] [T3]

Practical use means the reviewed text assigns or demonstrates work with the tool; an option does not establish that a student selected it. Twenty course families lack sufficiently explicit practical / listed platform evidence. No adoption of Supabase or PGlite was established in this archive.

Main conclusion

MySQL and Access have substantial documented footprints in the reviewed courses. Keep meaningful compatibility routes while using one principal SQL foundation. The choice of PostgreSQL is a design recommendation, rather than the dominant platform in this sample.

The proposed technology pathway

Recommendation: make **PostgreSQL** the SQL foundation, with **PGlite** for immediate browser practice and **Supabase** for the hosted project. Use **Power BI** as the main reporting studio, **Tableau** as an alternative, and **Lucidchart** for modeling with a supported alternative. Keep the existing **Microsoft Access** route; develop and test a matching **MySQL** route.

Layer	Recommended role	Condition for adoption
SQL	Main skill and reasoning language.	Portable core examples with PostgreSQL-specific syntax labeled.
PGlite	Immediate embedded practice.	Planned. Implement and test persistence, reset, accessibility and known-answer checks.
Supabase	Hosted project and operational controls.	Classroom onboarding, export/restore, permission tests and a supported exit route; no guaranteed free service claim.
Power BI	Principal BI studio in Chapter 14.	Windows authoring access or a supported institutional virtual lab.

Layer	Recommended role	Condition for adoption
Tableau	Equivalent alternate visualization studio.	Same data, measures and rubric; avoid two required BI tools in the standard route.
Lucidchart	Diagramming example.	Provide a supported alternative; no presumed free-tier entitlement.
Access / MySQL	Compatibility routes. Access is a route in use (labs and an appendix); MySQL is a planned route.	Test the MySQL route on the shared cases, with equivalent core examples and instructor setup; not every student must learn both.

PGlite executes locally in the browser; Supabase is hosted PostgreSQL. Export / import must be taught explicitly. Power BI Desktop requires Windows or an institutional virtual lab. “Browser-first SQL practice with a hosted project” accurately describes the intended path; integration, access, licensing, and maintenance still need implementation decisions.

Companions and appendices

Setup, alternative routes and reference material support the tools each instructor chooses.

Companion	Purpose
PostgreSQL / PGlite / Supabase	Setup, workspace reset/persistence, portable SQL, explicit transfer, exports, restore and access tests.
MySQL / MySQL Workbench	A planned alternate route, to be tested with matching data, query results and dialect differences. An appendix-length syntax list alone is insufficient.
Microsoft Access	In use today. Tables, relationships, visual queries, forms, reports and an equivalent small application.
Lucidchart and alternatives	Crow’s Foot modeling, exports and review; diagrams.net/ERDPlus or Mermaid alternatives avoid a mandatory subscription.
Power BI and Tableau	A complete Power BI studio plus an equivalent Tableau reporting route; Mac access planning is explicit.
SQL reference and advanced database topics	Portable core syntax, dialect flags, triggers/procedures, deeper windows and normal forms.
Capstone, reviews and instructor resources	Milestones, tests, rubrics, solution files, slides, question banks and course-route maps.

Running projects

Running cases

Every Let's Build uses one guided case and every lab uses a second case, so students learn a skill once and then transfer it. The Grading Database and PetVax are the existing running cases. A small retail business is proposed as an alternative for evaluation; it is not a completed third case.

Guided case · in use

Grading Database

Courses, assignments and grades offer a familiar setting, although policy complexity still needs careful scoping.

What to test: Keep as the guided case with a small, stable schema.

Lab case · in use

PetVax veterinary clinic

Customers, pets, visits, services and payments give real relationships and operating KPIs.

What to test: Whether students relate to the setting.

Proposed alternative · for evaluation

Small retail business

Orders, products, customers and revenue suit the analytics, Power BI and star-schema chapters.

What to test: Whether the extra entities overload an introductory lab.

Learning and assessment

Each chapter uses the components listed in the overview. Here is what each one assesses.

Assessment that makes reasoning visible

Students predict and define before execution, test keys and totals during execution, and interpret the result afterward. Reading checks, guided practice, independent labs, cumulative reviews, and the final project serve distinct purposes.

Assessment	Purpose	Proposed evidence
Reading assessment tests	Check readiness and conceptual distinctions.	Short scenario, output-prediction and misconception items; separate formative feedback from graded evidence.
Let’s Build	Guided skill development.	Worked reasoning, expected results and checkpoints.
Interactive chapter labs	Independent transfer.	SQL, numeric answers, diagrams, files and short interpretations, appropriate to the chapter.
Checking AI-assisted work	Practice judging AI-assisted work, from Chapter 5 on.	Accept, revise or reject an AI-written query or answer, with a short written reason.
Project checkpoints	Stage the semester project.	Proposal and data plan at the Midterm Integration Studio, a working database in Chapter 11, and the final project in Chapter 16.
Cumulative tests / reviews	Integrate ideas across chapters.	Unfamiliar schema or changed business rule; explain and repair.
Final project	Demonstrate a complete usable solution.	Rebuildable database, validation log, report, decision memo and individual demonstration.

Assessment principle

Use known-result SQL and integrity tests where they fit; use human review for modeling, explanations, and business judgment. Evaluate unfamiliar variations and require a short individual explanation. Automated feedback on SQL and lab answers is a prototype in testing, and AI-assisted feedback is a development option that needs validation.

One problem. Three checkpoints. A defensible recommendation.

• **Question and requirements.** A scoped business problem, stakeholder, measure, source plan and acceptance criteria. Start in the opening chapters.

• **Design and working data.** An ERD, normalization rationale, implemented schema, synthetic data, import/reject log and integrity checks.

• **Evidence and action.** Tested SQL, reconciled reporting, essential access/recovery controls, a decision memo and an individual change task.

Proposed rubric	Weight
Problem framing and requirements	10%
Data model and normalization rationale	20%
Implementation and data quality	20%
SQL correctness and verification	20%
Reporting and business interpretation	15%
Reproducibility, controls and individual explanation	15%

Rubric weights are a proposal for faculty review, not a change to the live course. A compact submission includes schema.sql, seed data, queries.sql, an ERD, validation evidence, README and a short decision memo. A live link alone is insufficient because its data may change after submission.

References

[S1]

Author's syllabus archive and manifest

63 public course documents (syllabi, catalog descriptions and schedules) from 31 institutions, collected from institutional websites on October 1, 2026. Mixed course levels and dates; the original source of each document is linked in the course register. Archive available on request.

[S2]

Author's database-platform evidence model

October 17, 2026 audit of the same archive. Role-coded platform evidence (practical use, option, listed) for 54 selected course families; counts recalculated October 25, 2026. Available on request.

[S4]

Author's textbook adoption and contents audit

October 10, 2026 report: 60 named books and course texts in the archive, with school-level mentions. Historical mentions, not market shares. Available on request.

[P2]

Cengage: Database Systems, 15th edition ↗

Coronel and Morris. Official current listing checked October 25, 2026: 16 chapters in six parts, with Introduction to SQL (Chapter 4) and Advanced SQL (Chapter 5) before ER modeling; auto-graded database labs in MindTap.

[P3]

Cengage: Concepts of Database Management, 10th edition ↗

Friedrichsen, Ruffolo, Monk, Starks, Pratt and Last. Official publisher search results provide the nine-module outline and Access practice description.

[P4]

Pearson: Database Concepts, 10th edition ↗

Kroenke, Auer, Vandenberg and Yoder. Current publisher educator page; the uploaded PDF is the older ninth edition, not the tenth.

[P5]

Pearson: Modern Database Management, 14th edition ↗

Hoffer, Topi, Venkataraman and Bala. Official contents and description checked October 25, 2026: 13 chapters, SQL from Chapter 5, an analytics chapter and new State of AI sections.

[L1]

U.S. Bureau of Labor Statistics: Data Scientists ↗

Current Occupational Outlook Handbook read October 25, 2026. Projects 35% employment growth, 2025-2035; all occupations 3%.

[L2]

U.S. Bureau of Labor Statistics: Database Administrators and Architects ↗

Current Occupational Outlook Handbook read October 25, 2026. Architects 9%; administrators 0%; combined 4%, 2025-2035. Rounded percentages.

[T1]

PGlite: What is PGlite? ↗

Official documentation: embedded PostgreSQL compiled to WebAssembly; runs in the browser, with optional local persistence. It is not itself a hosted cloud database.

[T2]

Supabase: Database overview ↗

Official documentation: each project provides a PostgreSQL database. Hosted operations, accounts, permissions and product limits require a separate teaching plan.

[T3]

Microsoft: Get Power BI Desktop ↗

Current installation documentation, checked October 25, 2026. Windows desktop authoring creates an access requirement for Mac users.